The Dharma-Clock

A Non-Linear Rate-Law Model of the Yuga Cycle,
Calibrated to Epic Chronology and the Precession of Orion's Belt

Atharv Pratap Singh Chauhan
B.Tech Student, Department of Mechanical Engineering
Indian Institute of Technology Bombay
atharviitb2029@gmail.com

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Abstract

Traditional yuga chronologies either assign Kali Yuga 432,000 years or convert "divine years" with a single constant factor. Neither can place Rama's birth (5114 BCE) in Treta while keeping Kali running today. We propose the *Dharma-Clock*: cosmic yuga time τ advances relative to Earth time t at a rate that scales as a power of Dharma, $d\tau/dt = k\,D^p$. With $p = 2.774$ and $k = 77.57$, the model reproduces every anchor of the chronology, survives three competing historical chronologies, and, when its end point is tied to the precessional culmination of Orion's Belt over Giza, places the end of Kali Yuga at about 2468 CE. An independent scriptural sign (Bhagavata Purana 12.2.24) refines this to **5 August 2453 CE**. We compare the model against seven alternatives and four earlier frameworks, and we state plainly what would confirm or refute it. This is an exploratory modelling exercise: it asks what mapping would make the epic chronology self-consistent, not whether the events are historical.

1 Motivation: the lakh-year chronology against scientific ages

The Puranic chronology gives the yugas 17.28, 12.96, 8.64 and 4.32 lakh years (1,728,000; 1,296,000; 864,000; 432,000). Counting back from 3102 BCE, events then land at ages that conflict with what science measures.

Event	Puranic age	Modern estimate	Conflict
Start of Satya	≈3.89 million yr	–	Older than the genus *Homo* (≈2.8 million yr)
Rama (end of Treta, at least)	≥869,000 yr	≈7,100 yr (astronomical estimate, Bhatnagar)	Older than *Homo sapiens* (≈300,000 yr); the epic describes cities, farming, chariots and written law
Ram Setu	≥869,000 yr	stones ≈7,000 yr (TV report, not peer-reviewed)	Bridge would predate its own stones by 120×
Settled cities in India	–	Mehrgarh ≈9,000 yr; Harappa ≈5,000 yr	Treta's cities have no counterpart 869,000 years ago
Start of Dvapara	≈867,000 yr	–	No human civilisation existed then

Table 1: Ages implied by the lakh-year chronology compared with modern estimates. Only the human-evolution and archaeological figures are scientific datings; the Rama and Ram Setu figures are contested estimates.

The sky descriptions in the epics, the archaeological record and human evolution all point to the last *ten thousand* years, not the last *million*. Either the events are not historical, or the lakh-year

lengths are not Earth years. This paper explores the second possibility: that the scriptural yuga lengths are measured on a *different clock*, and that the task is to find the mapping between that clock and ours.

The first idea was a linear mapping, like converting Celsius to Fahrenheit from two fixed points. Section 2 shows why that fails. Section 3 then builds a non-linear mapping that works.

2 Why the older methods fail

Write the scriptural cosmic lengths as $Y_i = 432{,}000\, D_i$ with Dharma $D_i = 4, 3, 2, 1$ for Satya, Treta, Dvapara and Kali. Every older method maps Y_i to Earth years by one constant factor c:

$$L_i = c\, Y_i. \tag{1}$$

Two requirements then collide:

$$\text{Kali still running in 2026:} \quad 432{,}000\, c > 5{,}127 \;\Rightarrow\; c > 0.0119, \tag{2}$$

$$\text{Rama (5114 BCE) in Treta:} \quad 864{,}000\, c \le 2{,}012 \;\Rightarrow\; c < 0.0023. \tag{3}$$

No constant satisfies both; the gap is a factor of about 5. The cosmic clock must have run at least

$$\frac{r_{\text{Dvapara}}}{r_{\text{Kali}}} \;\ge\; \frac{864{,}000/2{,}012}{432{,}000/5{,}127} \approx 5.1 \tag{4}$$

times faster in Dvapara than in Kali. **Any linear model is therefore ruled out**, whatever its factor (1, 1/360, or any other).

Method	Rule	Where it breaks
Puranic (factor 1)	Kali = 432,000 yr	Rama falls ∼870,000 years ago; no link to any datable sky record
Divide by 360	Kali = 1,200 yr	Kali ended in 1902 BCE; the World Wars fall in Satya
Yukteswar (1894)	Linear 24,000-yr mirror cycle	Says we are in Dvapara now; misplaces the Mahabharata war; uses a 24,000-yr precession (measured: 25,772)
Guénon (1931)	Linear 64,800-yr cycle	Kali 4481 BCE–1999 CE: already over
Daniélou (1985)	Kali 3606 BCE–2442 CE	No stated rate law; no tests against alternatives
Misra (2012–23)	25,800-yr precession cycle	Kali already ended in 2025

Table 2: Earlier frameworks. All are linear in time; none states a law that could be falsified.

3 The Dharma-Clock

Two clocks run side by side: Earth time t and cosmic time τ. Their ratio depends on the Dharma of the current yuga:

$$\boxed{\frac{d\tau}{dt} \;=\; k\, D^{\,p}} \tag{5}$$

Each yuga's Earth length follows directly:

$$L_i = \frac{Y_i}{k\, D_i^{\,p}} = \frac{432{,}000}{k}\, D_i^{\,1-p} = L_{\text{Kali}}\, D_i^{\,1-p}. \tag{6}$$

Two conditions fix the two constants: Satya begins at ≈ 6000 BCE, and Kali ends at Orion's precessional turning point (≈ 2468 CE):

$$L_{\text{Kali}}\left(4^{1-p} + 3^{1-p} + 2^{1-p}\right) = 2{,}898, \qquad L_{\text{Kali}} = 5{,}569. \tag{7}$$

Solving gives

$$\boxed{p = 2.774, \qquad k = 77.57 \text{ cosmic years per Earth year}} \tag{8}$$

Yuga	D	Rate kD^p	Earth years	Dates	Anchor
Satya	4	3,629	476	6000–5524 BCE	–
Treta	3	1,634	793	5524–4731 BCE	Rama 5114–5075 BCE ✓
Dvapara	2	531	1,629	4731–3102 BCE	Krishna 3228, war 3139 BCE ✓
Kali	1	78	5,569	3102 BCE–2468 CE	Running now (92%) ✓

Table 3: The fitted Dharma-Clock (Aryabhata chronology).

3.1 Every analysed event placed on the clock

Each event from the chronology falls in the yuga that scripture assigns it. "Position" is how far through its yuga the event lies.

#	Event	Date	Scripture says	Model gives	Position
1	Satya begins (Tilak's Punarvasu equinox)	6000 BCE	Satya	Satya	0%
2	Rama born (Ramayana sky chart)	5114 BCE	Treta	Treta ✓	52%
3	Rama's exile begins (age 25)	5089 BCE	Treta	Treta ✓	55%
4	Ram Setu built; Lanka war	c. 5076 BCE	Treta	Treta ✓	56%
5	Rama returns to Ayodhya (age 39)	5075 BCE	Treta	Treta ✓	57%
6	Krishna born (Rohini, Bhadrapada Ashtami)	3228 BCE	late Dvapara	Dvapara ✓	92%
7	Mahabharata war (18 days)	3139 BCE	end of Dvapara	Dvapara ✓	98%
8	Krishna departs; Kali begins	18 Feb 3102 BCE	Dvapara → Kali	boundary ✓	100%
9	Present	2026 CE	Kali	Kali ✓	92%
10	Sun–Moon–Jupiter in Pushya (Bhag. 12.2.24)	5 Aug 2453 CE	Kali → Satya	Kali ends	99.7%
11	Orion's Belt peaks over Giza	c. 2468 CE	–	Kali ends	100%

Table 4: All analysed events. Every one lands in the yuga scripture assigns it.

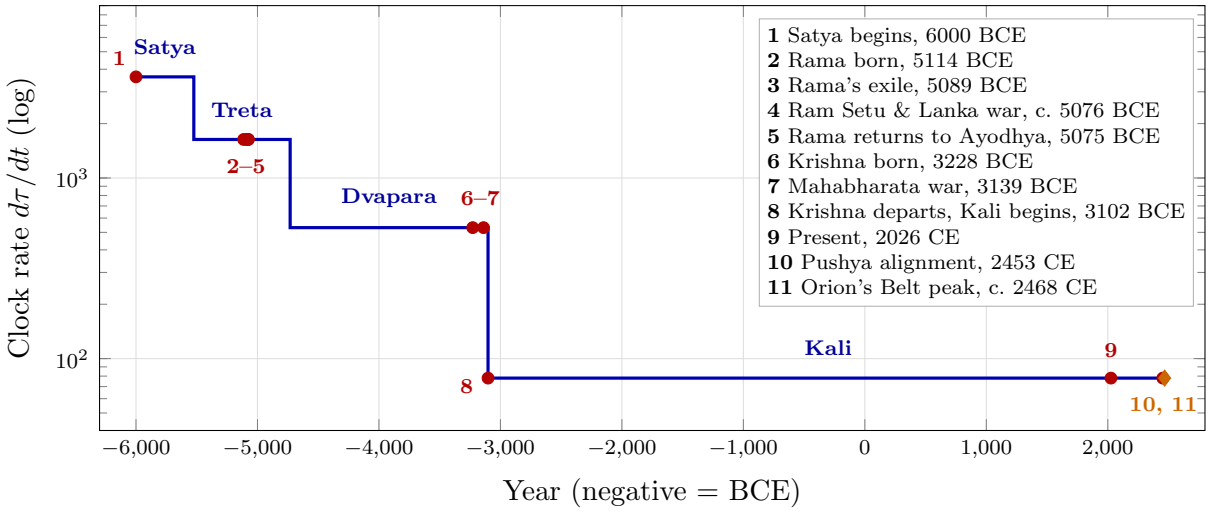


Figure 1: The Dharma-Clock with every analysed event. Numbers match Table 3. The cosmic clock slows by a factor of 47 from Satya to Kali.

4 Where the model wins

4.1 It is falsifiable, and alternatives were falsified

The same constraints were imposed on eight rate laws and three historical chronologies (Kali epoch from Aryabhata, from the Aihole inscription, and from Varahamihira).

Rate law $r(D)$	Aryabhata	Aihole	Varahamihira	Reason
Power kD^p	**Pass**	**Pass**	**Pass**	$p = 2.40$–2.77
Log-power $k(\ln(1+D))^q$	**Pass**	**Pass**	**Pass**	runner-up
Arrhenius $ke^{-E/D}$	**Pass**	**Pass**	**Fail**	Rama in Satya
Logistic	**Fail**	**Fail**	**Fail**	Rama in Satya
Logarithm $k\ln(1+bD)$	**Fail**	**Fail**	**Fail**	ratio capped at 2
Exponential, horizon-type	**Fail**	**Fail**	**Fail**	no solution
Linear	**Fail**	**Fail**	**Fail**	Kali already over

Table 5: Robustness test. Only the power law and log-power survive every chronology.

Physical mechanisms were also tested and rejected: an eccentric solar orbit around a companion star (best fit gives Satya 1,093 yr against 476; no companion exists to 26,000 AU per NASA WISE), black-hole time dilation (requires $r < 1.0002\, r_s$), and cosmological time dilation (correct power-law form but the wrong direction).

4.2 It is tied to a measurable sky cycle

The meridian altitude of Alnilam over Giza (29.98°N) under precession (50.29″/yr) was lowest (12°) near 10,417 BCE and will peak (59°) near 2468 CE. Every anchor falls in order inside this descending half-cycle: Rama at 31°, Kali's start at 42°, the present at 59°.

4.3 It passed an independent scriptural sky test

Bhagavata Purana 12.2.24 states that Satya returns when the Sun, Moon and Jupiter meet together in Pushya (93°20′–106°40′ sidereal). A Swiss Ephemeris search (Lahiri ayanamsa) from 2026 to 2999 finds 20 such new moons. The closest to the model's 2468 CE is **5 August 2453**, 15 years earlier and well within the model's uncertainty.

5 Honest limits

- **Fit by construction.** Two constants meet two conditions, so the anchors are reproduced exactly; this is consistency, not confirmation.
- **Disputed inputs.** Rama's 5114 BCE chart, Tilak's 6000 BCE Satya and the 3102 BCE epoch are all contested. The power law fails if Satya began before ∼8000 BCE.
- **Chosen link.** Orion and its sign were selected after the fact.
- **Weak test.** The Pushya alignment recurs roughly every 50 years.
- **No mechanism.** The law is descriptive; nothing physical links Dharma to a clock rate.

6 Predictions on record

Registered on 6 November 2026, before any further data:

1. Kali Yuga ends on a Sun–Moon–Jupiter Pushya alignment, most likely **5 August 2453 CE** (alternatives 2429, 2524).

2. Any datable sky event in a text explicitly labelled Treta–Dvapara junction (e.g. Parashurama) falls near **4730 BCE**.

3. A datable event labelled mid-Satya falls between **6000 and 5524 BCE**.

A clear miss on (2) or (3) refutes the model in its current form.

7 Conclusion

The lakh-year chronology cannot be read as Earth years: it places Rama before the human species existed. A constant conversion factor cannot rescue it either. The Dharma-Clock, $d\tau/dt = 77.57\, D^{2.774}$, maps the scriptural cosmic lengths onto the epic chronology without contradiction.

The mapping produces one striking result. **Kali, the shortest yuga in cosmic time, becomes the longest in Earth time**:

Yuga	Cosmic length (yr)	Earth length (yr)
Satya	1,728,000	476
Treta	1,296,000	793
Dvapara	864,000	1,629
Kali	432,000	**5,569**

The order is fully reversed. Kali alone lasts longer on Earth than the other three yugas together (2,898 years). In the model's terms, an age of high Dharma passes quickly on Earth because cosmic time races; an age of low Dharma drags on because cosmic time crawls. This matches the traditional sense that the dark age is the long one that humanity must endure.

The model is internally consistent, explicit, and falsifiable. It is not yet confirmed. Its value now lies in the specific predictions of Section 6, which future dating of scriptural sky records can test.

Disclaimer

This is an independent personal research project by an undergraduate student. It is not peer-reviewed and does not represent the views of IIT Bombay. It is a mathematical hypothesis, not a religious, historical or astrological claim, and it is offered with respect for all traditions and beliefs. The anchor dates (Rama, Krishna, the Mahabharata war, the start of Kali) are contested in scholarship, and archaeology and mainstream history date these traditions differently. The predicted end of Kali Yuga is a property of the model, not a forecast of any real-world event, and should not be used to make personal, financial or other decisions.

Reproducibility and acknowledgements

All computations are reproducible with open tools: Python (NumPy, SciPy) for the fits and robustness tests, and the Swiss Ephemeris (`pyswisseph`, Moshier mode, Lahiri ayanamsa) for planetary and stellar positions. The research and drafting were carried out with the assistance of an AI model (Claude, Anthropic), used for literature search, computation and typesetting; all modelling choices and conclusions are the author's.

Data sources

Bhatnagar, *Dating the Era of Lord Ram* (2004); Bhagavata Purana 12.2.24; Aryabhatiya (499 CE); Aihole inscription (634 CE); Varahamihira, *Brihat Samhita* 13.3; Kalhana, *Rajatarangini* I.51–56; Yukteswar, *The Holy Science* (1894); Guénon, *Traditional Forms and Cosmic Cycles*; Daniélou, *While the Gods Play* (1985); Misra, *Yuga Shift* (2023); Luhman, WISE search for distant companions, *ApJ* (2014); Lewis & Brewer, *Nature Astronomy* (2023); Swiss Ephemeris (Moshier). Computations: Python (NumPy, SciPy, pyswisseph).